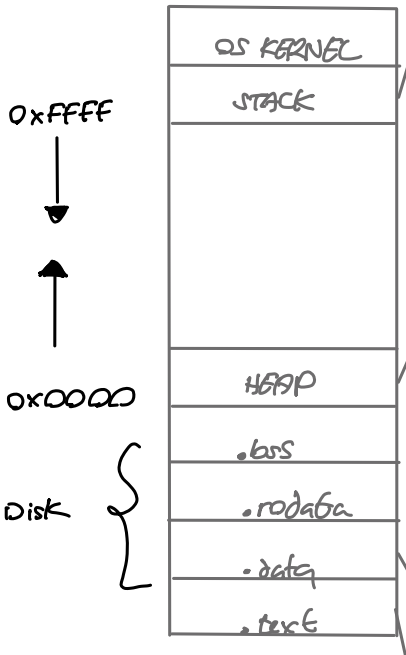
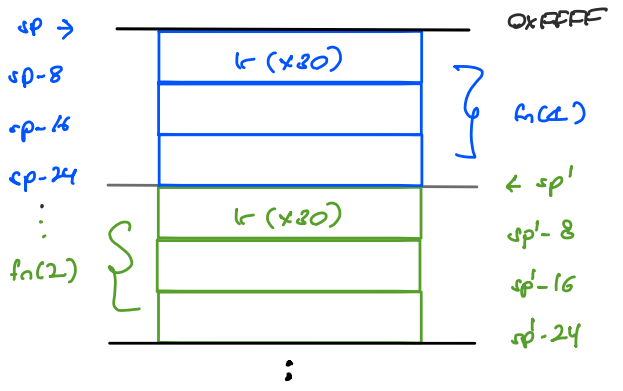
Program Memory Space



CALL FRAMES



Allocating Data structures: / Dynamic Mem / Memory leaks

- stack allocation : stack pointer ~ compiler / CPU
- heap allocation : managed by allocator *

Uninitialized Data — .bss

global or static vars — msg // char msg[100];

x explicit value — x // int x;

Initialized Data — .data:

.rodata ~ non modifiable — msg = "Hello"

.data ~ initialized writable data — x = 5

// global or static vars with explicit value

ARM 64 Instructions

.txt:

fn1:
- sub sp,sp,25
- str Lr,[sp,16]
- Mov x0, #5
- bL fn(2)

fn(2):
- sub sp,sp,25
- str Lr,[sp,16]
- :

Application / Runtime Memory Management

Manual Allocator

- allocate + dealloc pairs
- uncomplicated and buggy
- In C code

malloc() + free()

```
int *p = malloc(sizeof(int)*10))
free(p)
```

Automatic Allocator

- allocate and forget
- garbage collection

(Ref. counting) (Mark sweep) (stop+copy)

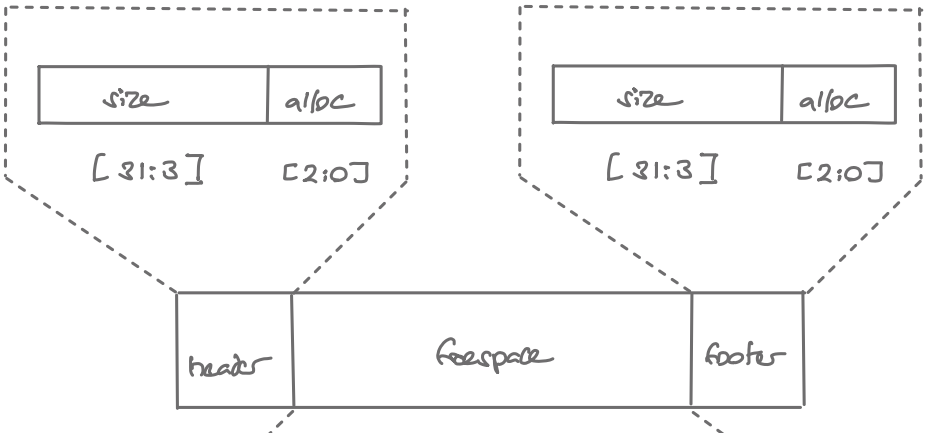
X traversal | traversal + cleanup | traversal + relocation

Reachability Graph, Traversal: BFS + DFS

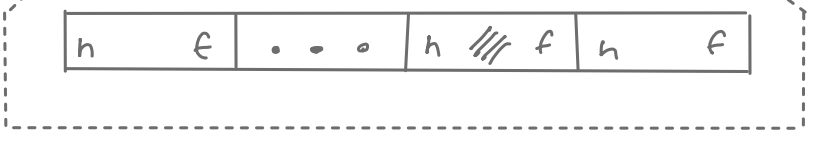
Ptr Reversal, TRI COLOR ABSTRACTION, Back table

Heap Allocator

- ① Block Representation



* block sizes aligned on 8B, 16B, etc..



- internal fragmentation
- external fragmentation

- (2). allocation strategies

first fit

```
p = heap start

if at heap end
    fail

if (block is free) &
   (blocksize ≥ alloc_size)
    allocate block
else
    p += block size
```

next fit

```
p = last_ptr

if (block is free) &
   (blocksize ≥ alloc_size)
    allocate block
else
    p += block size

if at heap end
    p = heap_start

if p = last_ptr
    fail
```

best fit

```
best = NULL;
best size = 0;

p = heap start;

if at heap end
    stop;

if (block is free) &
   (blocksize ≥ alloc_size)

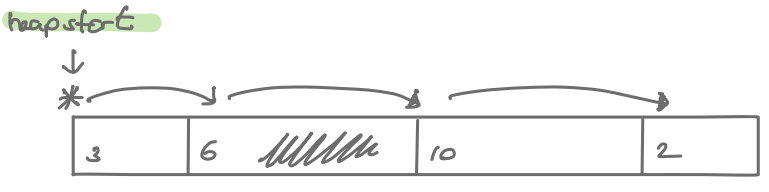
    if (blocksize < best_size)
        best_size = block size

if (best != NULL) allocate
else fail
```

- (3). search / traverse

Implicit Free List

- link by size



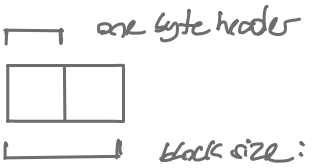
Explicit Free List

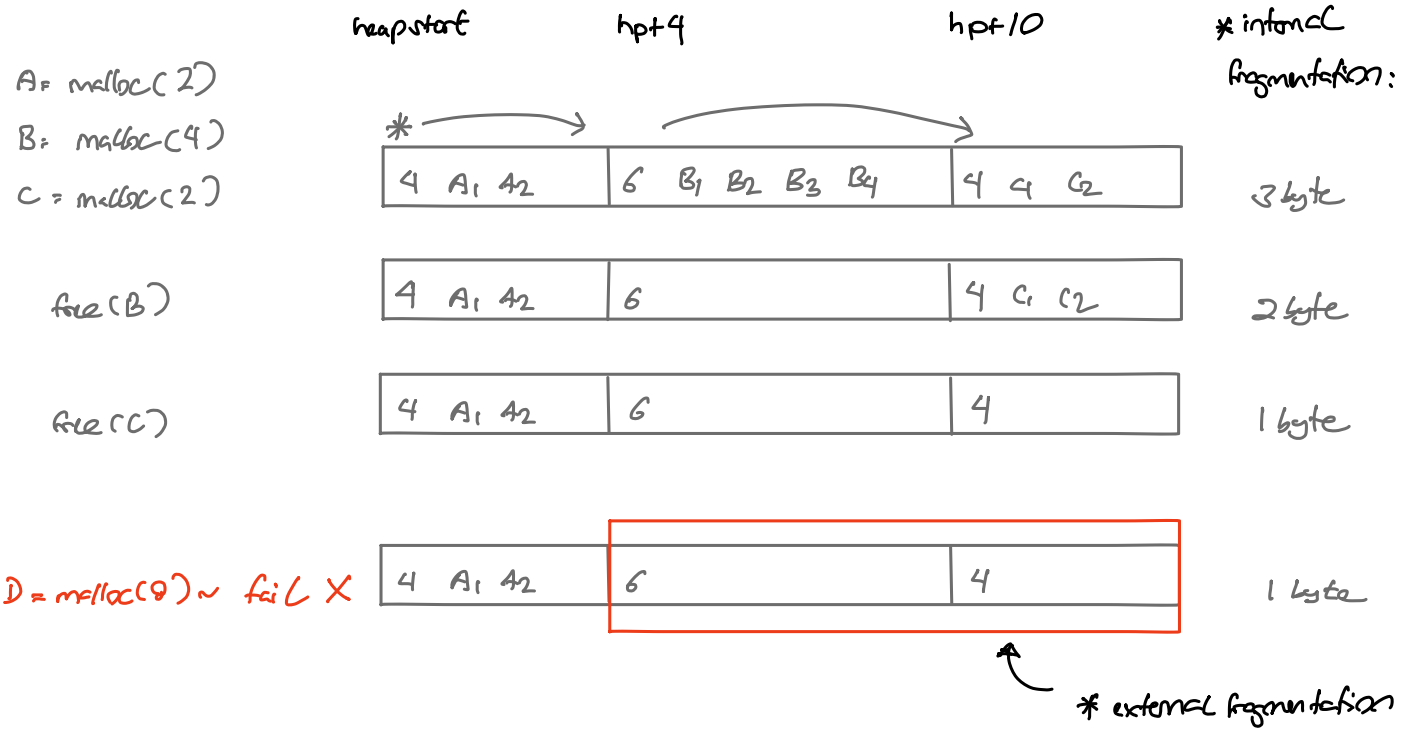
- link by ptr
- * only free blocks



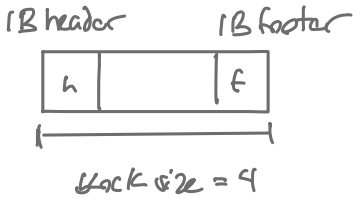
- ③ Coalescing :

- mem allocator : operating on 16 byte mem segment.
- allocation strategy : Implicit FreeList + Next Fit





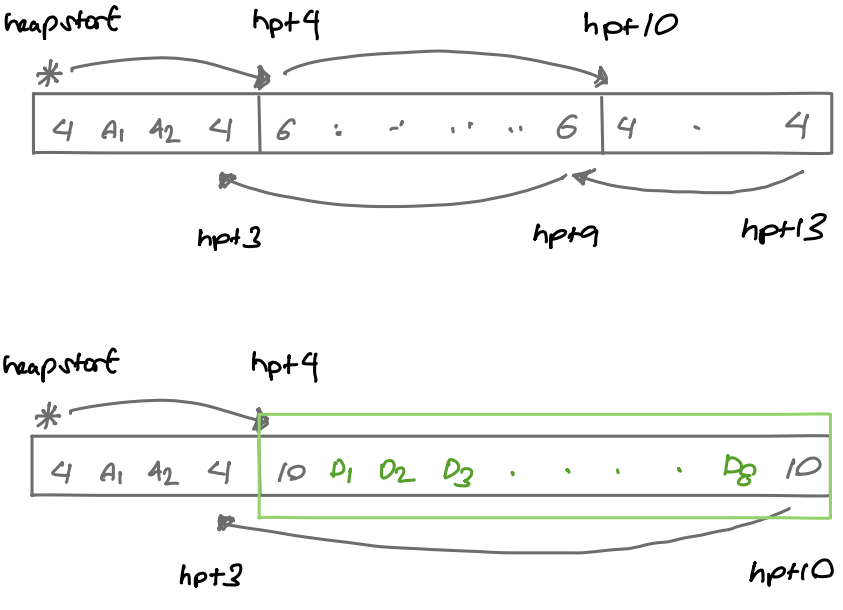
Bidirectional Implicit Free List



start with block N
free size = K

if (N+1) = free
free size += M

if (N-1) = free
free size += N



D = malloc(8) ~ succeed ✓